

- Q.17 Explain in brief film wise condensation. (CO3)
- Q.18 Explain in brief the temperature length curve for condenser with the help of diagram. (CO3)
- Q.19 Discuss in brief about boiling point elevation. (CO3)
- Q.20 Write short on multiple effect evaporator. (CO4)
- Q.21 Explain in brief the working of long tube vertical evaporator. (CO4)
- Q.22 Discuss in brief the different feeding arrangement in evaporator. (CO4)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Write short note on :- (CO2)
- Extended surface equipment
 - Individual and over all heat transfer coefficient
- Q.24 Derive an Expression for LMTD in case of Co-current flow by giving suitable assumptions. Also draw neat diagram. (CO1)
- Q.25 Explain in detail construction and working of Cochran Boiler. (CO3)

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Roll No.

5th Sem/ Chemical Subject : Heat Transfer Operations-II

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Dropwise condensation usually occurs on. (CO3)
- Rough surface
 - Curved surface
 - Shiny surface
 - Coated surface (Coating with oil, Teflon)
- Q.2 In a heat Exchanger, the rate of heat transfer from the hot fluid to the cold fluid (CO1)
- Directly proportional to the area and the LMTD
 - Directly Proportional to LMTD and inversely Proportional to the area
 - Varies as square of the area
 - None of these

- Q.3 Kg of water evaporated per hour in a evaporation process is called (CO4)
- Evaporator economy
 - Evaporator capacity
 - Evaporator efficiency
 - None of these
- Q.4 In which of the following water flows through shell of the boiler. (CO3)
- Fire tube Boiler
 - Water tube boiler
 - Both a & b
 - None of these
- Q.5 Condenser converts _____. (CO3)
- Solid feed to liquid phase product
 - Liquid feed to vapour phase product
 - Liquid feed to a solid phase product
 - Vapour feed to a liquid phase product
- Q.6 In which of the following flow, hot fluid and cold fluid flow in same direction. (CO1)
- Laminar Flow
 - Co-Current Flow
 - Counter current flow
 - Turbulent flow

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Write full form of LMTD. (CO1)
- Q.8 Name any one type of fin used in heat exchanger. (CO2)
- Q.9 Name any one water tube boiler. (CO3)
- Q.10 Define heat exchanger. (CO1)
- Q.11 Name any one type of condenser. (CO3)
- Q.12 Name any one type of evaporator. (CO4)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Discuss the various type of flow arrangement in heat exchanger. (CO1)
- Q.14 Explain in brief the working of double pipe heat exchanger. (CO1)
- Q.15 Define fouling and explain in brief how fouling affect the working of heat exchanger. (CO2)
- Q.16 Differentiate between water tube and fire tube boiler. (CO3)